

3. (a) The first law of thermodynamics is given by:

$$\Delta U = Q - W$$

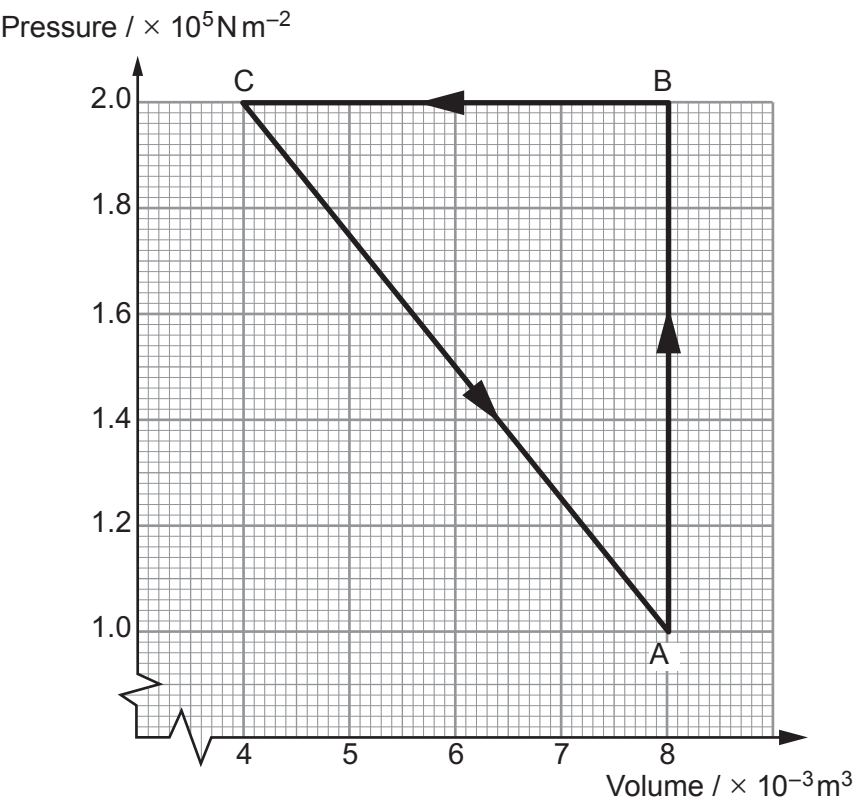
State what is represented by: [3]

ΔU

Q

W

(b) A fixed mass of gas is taken around the closed cycle $A \rightarrow B \rightarrow C \rightarrow A$.



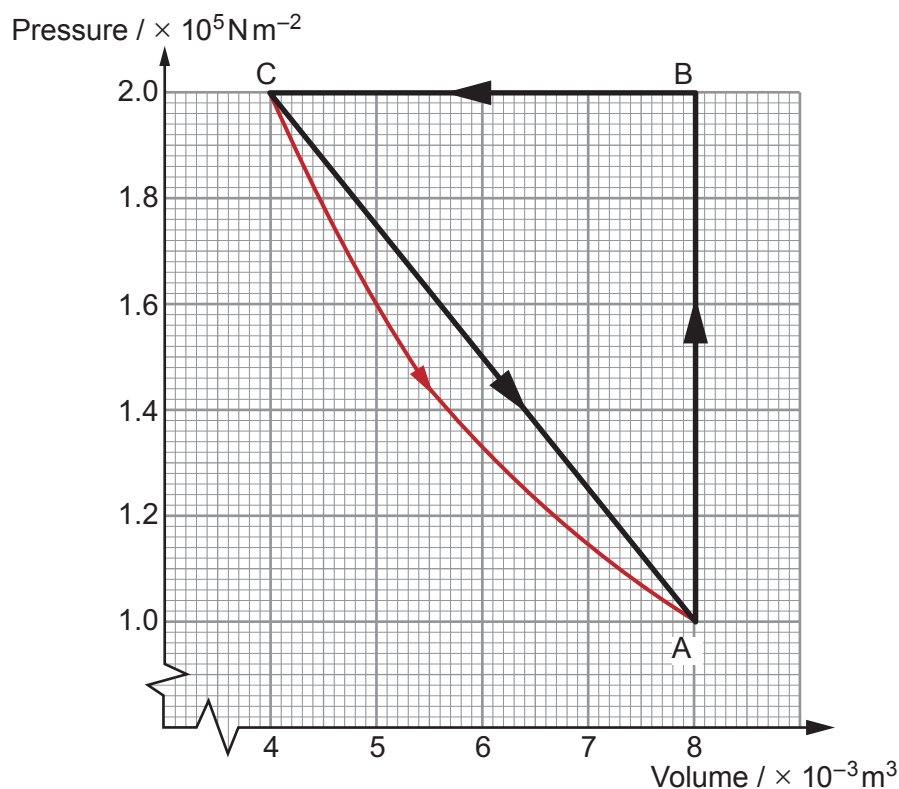
(i) Complete the table by describing each process in terms of pressure and volume. The description for process $A \rightarrow B$ is already inserted. For each process state if any work is done, and if so indicate if it is done *on* or *by* the gas. *No calculations are required.* [5]

Process	Description of process	Work done on / by gas (if any)
$A \rightarrow B$	Increase in pressure at constant volume	
$B \rightarrow C$		
$C \rightarrow A$		

- (ii) Determine the net work done on the gas during the entire cycle.

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- (c) The same mass of gas is taken around the cycle $A \rightarrow B \rightarrow C \rightarrow A$ for a second time. For this cycle the temperature is kept constant between C and A. The new path from C to A is shown in red on the graph below, together with the original path.



- (i) Show that the temperature is constant along the new path from C to A.

[3]

- (ii) Estimate the difference in the net work done in the two cycles.

[2]

(b) A container of fixed volume contains oxygen gas at a temperature of 293 K.

- (i) Five oxygen molecules have speeds 400, 425, 450, 550 and 625 m s⁻¹. Determine their rms speed. [2]

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- (ii) Using an appropriate calculation, determine whether or not the rms speed calculated in part (b)(i) is consistent with the expected rms speed of the molecules of the gas at this temperature. (Relative molecular mass of oxygen gas = 32.) [3]

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- (iii) If the gas in the container is heated and the pressure of the gas increases by 20% of its initial value, determine the rms speed of the molecules of the gas. [2]

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2. Kinetic theory for an ideal gas gives an expression that can be written as:

$$pV = \frac{1}{3} nN_A m \overline{c^2}$$

- (a) State the meaning of the terms: [2]

(i) m

(ii) $\overline{c^2}$

- (b) Defining both symbols, explain what quantity is given by nN_A . [3]

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- (c) The product of pressure and volume for an ideal gas may also be expressed as $pV = nRT$. Show in clear steps that the total translational kinetic energy of one mole of the gas is $\frac{3}{2} RT$. [4]

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- (d) An ideal monatomic gas, initially at a pressure of 115 kPa and temperature of 294 K, expands at constant pressure from a volume of $2.20 \times 10^{-3} \text{ m}^3$ to $2.60 \times 10^{-3} \text{ m}^3$. Calculate the change in the internal energy of the gas. [3]

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3. (a) Give the definition of the *specific heat capacity* of a material. [1]

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- (b) An insulating flask contains $0.6 \times 10^{-3} \text{ m}^3$ of water at 19.5°C . A volume $1.6 \times 10^{-3} \text{ m}^3$ of boiling water at 100.0°C is poured into the flask.

(specific heat capacity of water, $c = 4\,200 \text{ J kg}^{-1}^\circ\text{C}^{-1}$; density of water, $\rho = 1\,000 \text{ kg m}^{-3}$)

- (i) Determine the final temperature of the water in the flask. [3]

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- (ii) Calculate the heat lost by the boiling water. [3]

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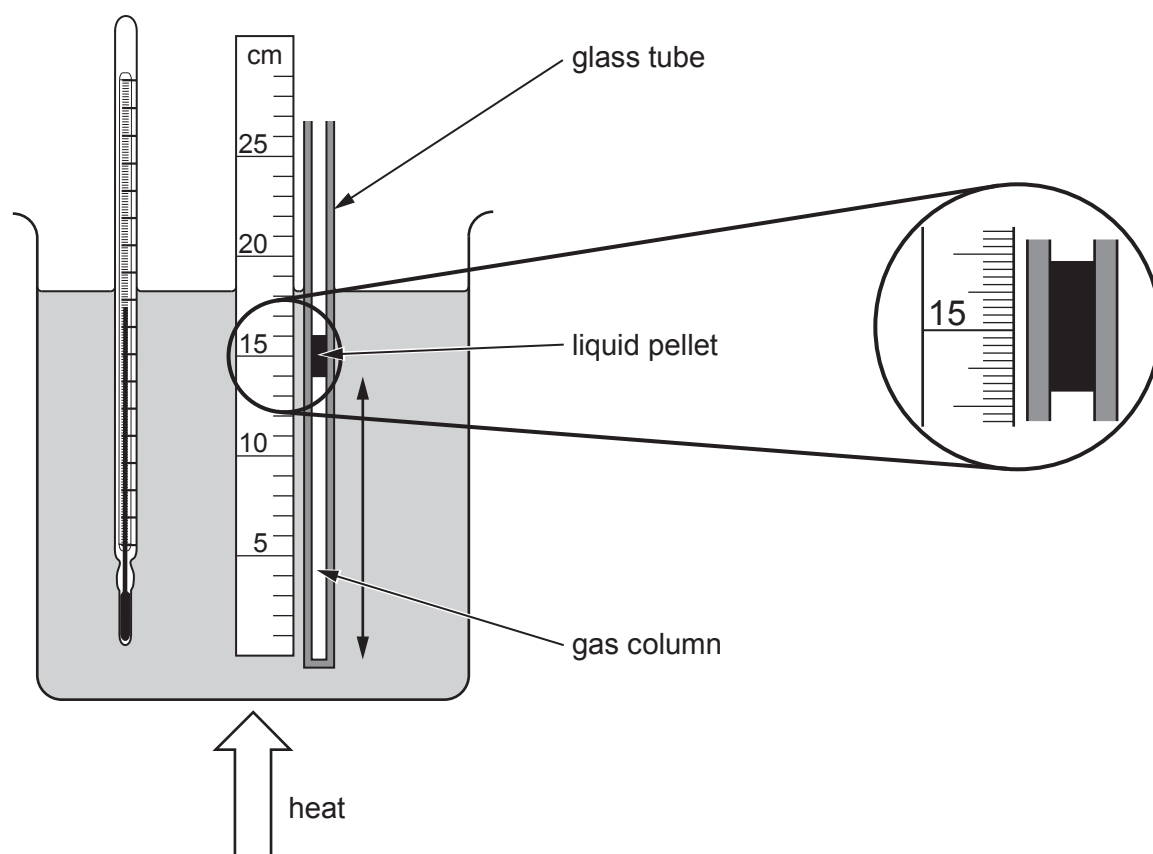
- (iii) Justify the statement that no work is done on/by the boiling water as it cools. No further calculations are needed. [1]

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SECTION A*Answer all questions.*

1. A student performs an experiment to estimate the Celsius temperature of absolute zero.



The length, l , of the gas column and its temperature, θ , are measured at atmospheric pressure, 1.01×10^5 Pa. The temperature of the water is initially 0.0°C . It is then increased to 80.0°C . Readings are taken at 20.0°C intervals. The scale used to measure the length gives an uncertainty of ± 0.1 cm at the top of the column and an uncertainty of ± 0.1 cm at the bottom.

Values of θ and l are recorded in the table below.

$\theta / ^\circ\text{C}$	l / cm
0.0	11.5
20.0	12.5
40.0	13.2
60.0	14.2
80.0	15.0



- (a) Justify the number of significant figures used to record the length, l . [2]

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- (b) The glass tube in which the gas is trapped has an internal diameter of (1.5 ± 0.1) mm.

- (i) Calculate the volume of the gas at 0.0°C . [2]

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- (ii) Show that the percentage uncertainty in this volume, V , is approximately 15%. [3]

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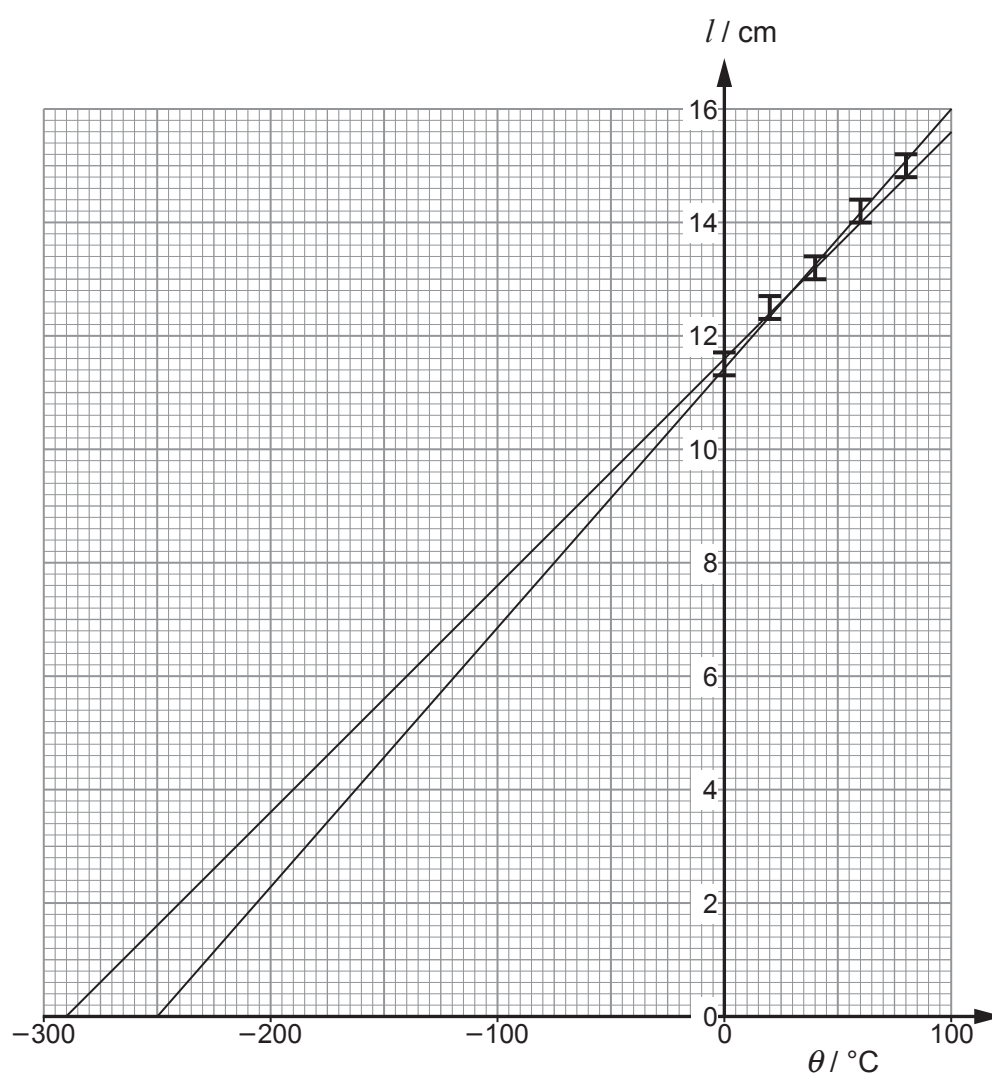
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- (c) The length, l , is plotted against the temperature, θ . Lines of maximum and minimum gradient consistent with the error bars have been drawn.



- (i) Use the graph to estimate the temperature of absolute zero in $^\circ\text{C}$, together with its **absolute** uncertainty. [2]

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(ii) Justify the statement that *the results are consistent with the ideal gas equation*. [4]

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(d) State what happens to the behaviour of the gas molecules as the temperature approaches absolute zero. [1]

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(e) The scale resolution of the ruler affects the final uncertainty in the value of absolute zero. Identify **two** other factors that could affect the accuracy. [2]

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2. A balloon of volume $1.60 \times 10^{-3} \text{ m}^3$ contains helium at a pressure of $1.20 \times 10^5 \text{ Pa}$ and a temperature of 293 K . (Relative molecular mass of helium = 4.)

(a) Determine:

- (i) the number of molecules in the balloon; [2]

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- (ii) the rms speed of the molecules. [2]

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- (b) The product of the pressure and the volume of the gas is given by $pV = nRT$ and also by $pV = \frac{1}{3}Nmc^2$.

- (i) Show that the mean kinetic energy of a gas molecule is $\frac{3}{2}kT$ where k is the Boltzmann constant. [4]

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(ii) Calculate the mean kinetic energy of a molecule in the gas. [1]

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(iii) Determine the internal energy of the gas in the balloon. [1]

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(c) (i) Explain, in terms of the molecules, the difference in the nature of the **internal energy** in an ideal gas and in a liquid. [3]

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(ii) State the temperature of a system for its internal energy to be a minimum. [1]

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SECTION AAnswer **all** questions.

1. The first law of thermodynamics may be written as:

$$\Delta U = Q - W$$

- (a) State the meaning of:

[3]

ΔU

Q

W

- (b) A closed cylinder fitted with a leak-proof piston as one end contains 0.060 mol of an ideal monatomic gas. The volume of the system is $1.4 \times 10^{-3} \text{ m}^3$ and the pressure of the gas is 100 kPa. Determine the temperature of the gas. [2]

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- (c) The gas is heated and expands at constant pressure to a volume of $2.0 \times 10^{-3} \text{ m}^3$. Determine:

- (i) the work done by the gas,

[2]

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- (ii) the increase in the internal energy of the gas,

[2]

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- (iii) the heat transferred to the gas.

[1]

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4. (a) Explain what is meant by specific heat capacity. [2]

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(b) A volume of $1.2 \times 10^{-3} \text{ m}^3$ of water is heated in a 3 kW kettle.

Specific heat capacity of water (and tea) = $4\,200 \text{ J kg}^{-1} \text{ K}^{-1}$

Specific heat capacity of milk = $3\,900 \text{ J kg}^{-1} \text{ K}^{-1}$

Density of water = $1.00 \times 10^3 \text{ kg m}^{-3}$

Density of milk = $1.03 \times 10^3 \text{ kg m}^{-3}$

(i) Determine the time it takes to increase the temperature of the water from 18°C to 100°C . [3]

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(ii) A volume of $0.25 \times 10^{-3} \text{ m}^3$ of the heated water is used to make tea and this is poured into a thermos flask. The temperature of the tea when in the flask is 95°C . Milk at 5°C is then poured into the flask. Determine whether $3.6 \times 10^{-5} \text{ m}^3$ of milk will lower the temperature of the mixture to the required temperature of 84°C . [4]

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7. (a) State **three** assumptions of the kinetic theory of ideal gases.

[3]

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- (b) A container of fixed volume contains oxygen gas. Five molecules in the gas have speeds 480, 521, 436, 445 and 503 m s^{-1} .

- (i) Determine the rms speed of these five molecules.

[2]

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- (ii) Determine the temperature of a sample of oxygen gas whose molecules have this rms speed. (Relative molecular mass of oxygen = 32.)

[3]

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- (c) Gareth states that when the temperature of the gas is doubled both the rms speed of the molecules and the pressure are doubled. Determine whether or not Gareth is correct.

[4]

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